Blood-Focused Dynamic Inversion Times for 3D LGE Imaging: Initial Patient Demonstration

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Synopsis

3D Late Gadolinium-Enhanced (LGE) imaging is used to assess scarring in patients with Atrial Fibrillation (AF). Acquiring image data during every cardiac cycle allows a reasonable total scan duration but exacerbates ghosting artefacts caused by variable heart rates, such as those of patients with AF. Dynamic Inversion Time (TI) methods improve image quality by modifying the TI for each cardiac cycle. We present the initial stage of a patient study to validate a recently-proposed blood-focused dynamic TI algorithm. No improvement was found in comparison to the original algorithm. Future comparisons will include more patients with high R-wave interval variability.

Introduction

Inversion-recovery 3D LGE imaging is a useful tool to map atrial scar distribution in patients with Atrial Fibrillation (AF). However, arrhythmia during the 3D scans in this patient population causes ghosting and poor myocardial nulling. Alternate-cycle inversion-recovery imaging [1, 2] reduces the sensitivity to heart rate variability but requires a longer scan duration unless an adaptation is made, which could potentially introduce other compromises.

Dynamic TI algorithms [3, 4] improve 3D LGE image quality when acquiring in every cardiac cycle, by updating TI on a beat-by-beat basis. The original algorithm [3] improved myocardial nulling by adjusting TI in “real time” in the current cycle of the sequence according to the previous cardiac cycle duration. While myocardial nulling is important, phase-encode ghosting from variations in the brighter blood signal remains troublesome in 3D imaging.

Reductions in blood ghosting, by targeting consistent blood signal across the TIs while maintaining low myocardial signal, were shown by simulations and phantom studies for a blood-focused dynamic TI algorithm [4]. Figure 1 compares the method of [4] with [3] in terms of blood Mz. Here we extend the work of [4] to a preliminary patient study.

Methods

We compared the performance of the blood-focused algorithm [4] with the original algorithm [3]. The blood-focused algorithm uses the full trigger interval and sequence timing history (including partial-saturation by acquisition) to model the blood and myocardium longitudinal magnetisation (Mz) evolution in real time during the sequence acquisition. It uses blood T1, myocardium T1 and an estimation of the mean RR interval to calculate the required TI in the current cycle to consistently achieve an unvarying target blood Mz, which is the blood Mz at the myocardium null time for the given mean interval.

3D LGE inversion-recovery scans were performed after 0.1mmol/kg gadolinium-based-contrast-agent (gadobutrol) administration in 13 patients with congenital heart disease, under local ethics approval. Each patient was scanned with two 3D LGE acquisitions, first the original dynamic TI algorithm [3] followed by the blood-focused algorithm [4] with otherwise identical imaging parameters, using single R-wave gating and the CLAWS phase-encode strategy in free respiration [5] on an Avanto 1.5T scanner (Siemens Healthineers, Erlangen Germany): segmented 3D spoiled gradient-echo, 32-36 slices, 1.5x1.5x4mm; reconstructed to 64-72 slices, 0.7x0.7x2mm. TI scout scans were performed before each 3D LGE acquisition, to estimate myocardium T1 for each algorithm. Blood T1 was measured prior to the blood-focused algorithm, using a single-slice short-axis MOLLI acquisition. All target blood Mz values were calculated as if the mean RR-interval was 1000ms.

To assess changes in blood ghosting artefact, signal was measured from regions positioned to detect phase-encode ghosting from blood in the heart (i.e. in-line with the heart in the right-left primary phase-encoding direction, Fig. 2) using the central 11 slices of the 3D imaging. As reference, mean intraventricular blood signal was measured for the same slices. To account for blood signal changes between the two 3D LGE scans, ghosting artefact mean was measured relative to blood signal mean.

Results

Figure 2 shows the centre slices for patient 5 as an example of image quality and signal intensity.

Measured ghost and blood signal means, with ghost-to-blood ratios, are shown in Fig. 3. No significant changes between the two algorithms (two tailed Student t-test of 13 pairs) were detected in mean ghosting ($p>0.4$) or ghost-to-blood ratio ($p>0.5$). The intraventricular mean blood signal was 72+/-31 for the original algorithm and 70+/-21 for blood-focused algorithm ($p>0.7$).

Figure 4 shows the ghost-to-blood ratio versus the R-wave interval variability for the 13 pairs of 3D LGE scans, where large R-R interval standard deviation represents high heart rate variability. The highest R-R interval variability as a percentage of the corresponding mean R-R interval was observed for patients 3 and 8, at 23%. There was no significant change in mean RR interval between the two 3D LGE scans ($p>0.7$).

Discussion

No significant difference was observed between the two algorithms. For each subject, variations in blood signal between the 3D LGE scans (arising from the target blood Mz selection mixed with gadolinium “wash-out”) may have influenced the blood ghosting-to-signal ratio. Clinical imaging required the original algorithm to be run first although randomised order was preferable.

Future studies are needed to increase the cohort size as differences in respiratory motion and associated residual PE ghosting between the two 3D scans were likely a major confounder in the ghosting assessment, even though ROIs were placed to avoid respiratory ghosting of the anterior chest wall. Further, more patients with AF are necessary.

Randomised scan acquisition order could reduce bias from blood T1 changes. The blood-based algorithm depends on an accurate Mz model, which depends on RF flip angle accuracy, making it important for RF flip angle mapping to be performed to assess the B1+ variation within the heart.

Myocardial ghosting possibly confounded the blood ghosting measurement, which could be reduced using a patient-specific mean interval for each target blood Mz calculation.

Conclusions

No improvement was found in this initial stage of an *in vivo* validation of the blood-based dynamic TI algorithm [4] for improved image quality in 3D LGE imaging. Assessment in a larger group is required, including more arrhythmia patients.

Acknowledgements

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References

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Figures

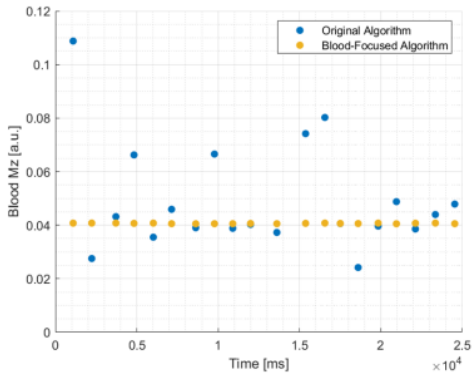

Figure 1: Example simulation of blood Mz at each TI for a given set of R-wave trigger intervals, demonstrating the difference between the blood-focused algorithm and the original algorithm. Blood Mz is plotted against elapsed time during the scan. The target blood Mz (0.046) depends on the estimated interval mean (812ms used here).

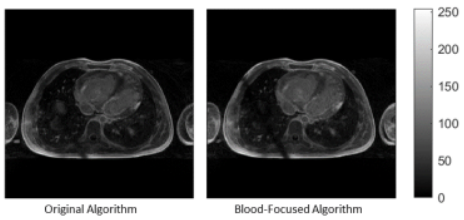

Figure 2: Example centre slice, for patient 5. In this case, blood signal is higher for the blood-focused algorithm images.

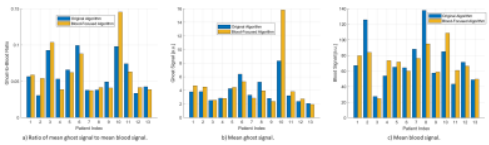

Figure 3: Metrics used to assess the two dynamic TI algorithms, for each patient: a) The ratio of mean ghost signal to mean blood signal (Ghost-to-Blood Ratio) for each 3D LGE scan, b) mean ghost signal within the select regions of interest, c) mean blood signal.

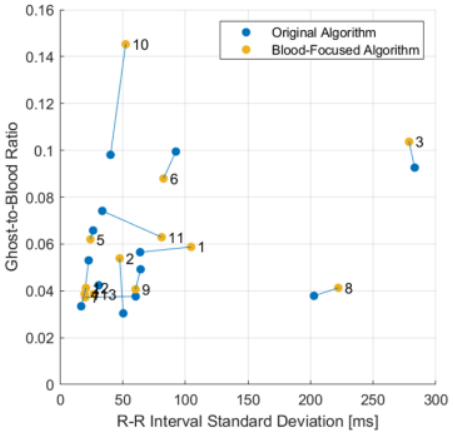

Figure 4: The ratio of mean ghost signal to mean blood signal for each patient, plotted against heart rate variability. Patients with arrhythmia would exhibit greater heart rate variability (i.e. larger R-wave trigger interval standard deviation) than those with a normal sinus rhythm. Each line indicates a pair of 3D LGE scans for an individual patient. Each pair is labelled with the corresponding patient index. Note two patients (8 and 3) exhibited high R-wave interval variability (e.g. R-R interval standard deviation > 200ms).